

CY 555: Introduction to Polymer Science & Technology

Course Code	CY 555
Course Name	Introduction to Polymer Science & Technology
L-T-P-C	3-0-0-3
Prerequisites	Undergraduate Level Chemistry or Teacher's Consent
Intended for	BS-CS/MSc/PhD
Distribution	Elective
Mutual Exclusion	-
Approval	33rd BoA (Jan 2020); 24th Senate (Feb 2020)
Revised	59th BoA (Apr 2025) - syllabus revised

Course Contents

- **Introduction to polymers:** History and recent developments, monomers, oligomers, polymers and their characteristics, classification and nomenclature of polymers, physical state of polymers, T_g, T_c, molecular weight and MWD, natural polymers. [6 Lectures]
- **Radical polymerization:** Mechanism, kinetics, chain transfer, autoacceleration, gel effect, copolymerization, reactivity ratios, composition of copolymers, living radical polymerization: ATRP and RAFT. [6 Lectures]
- **Ionic polymerization:** Mechanism and kinetics of cationic and anionic polymerization. [5 Lectures]
- **Stereospecific polymerization:** Stereoisomerism, complex catalyst polymerization. [2 Lectures]
- **Step growth polymerization:** Kinetics, step copolymerization, detailed methods for the preparation of polyesters, polyamides, polycarbonates etc., high performance polymers. [6 Lectures]
- **Techniques of polymerization:** Bulk, solution, emulsion, suspension, melt polycondensation, solution polycondensation, interfacial and gas phase. [3 Lectures]
- **Polymer characterization:** Molecular weight by GPC, light scattering, osmotic pressure etc., IR, UV, NMR, TGA, DSC, radiation scattering: SAXS, WAXS, DLS. [8 Lectures]
- **Processing, testing and applications of polymer materials:** Extrusion, molding, tensile, impact, flexural testing, adhesives, foam, polymer fibers, catalysis, environment care, medicine etc. [6 Lectures]

Text Books

1. V. R. Gowarikar, N. V. Viswanathan, J. Sreedhar, **Polymer Science**, New Age International, Wiley, 1995.
2. F. W. Billmeyer, **Textbook of Polymer Science**, 3rd Edition, Wiley, 2007.

References

1. G. Odian, **Principles of polymerization**, 4th Edition, Wiley, 2004.
2. P. C. Heimenz, T. P. Lodge, **Polymer Chemistry**, 2nd Edition, CRC press, 2007.
3. C. E. Carraher, **Seymour/Carraher's Polymer Chemistry**, 6th Edition, Marcel Dekker, Inc., 2003.
4. J. M. G. Cowie, **Polymers: Chemistry and Physics of Modern Materials**, 3rd Edition, CRC Press, 2007.
5. H. F. Mark, **Encyclopedia of Polymer Science and Technology**, 3rd Edition, Wiley, 2004.